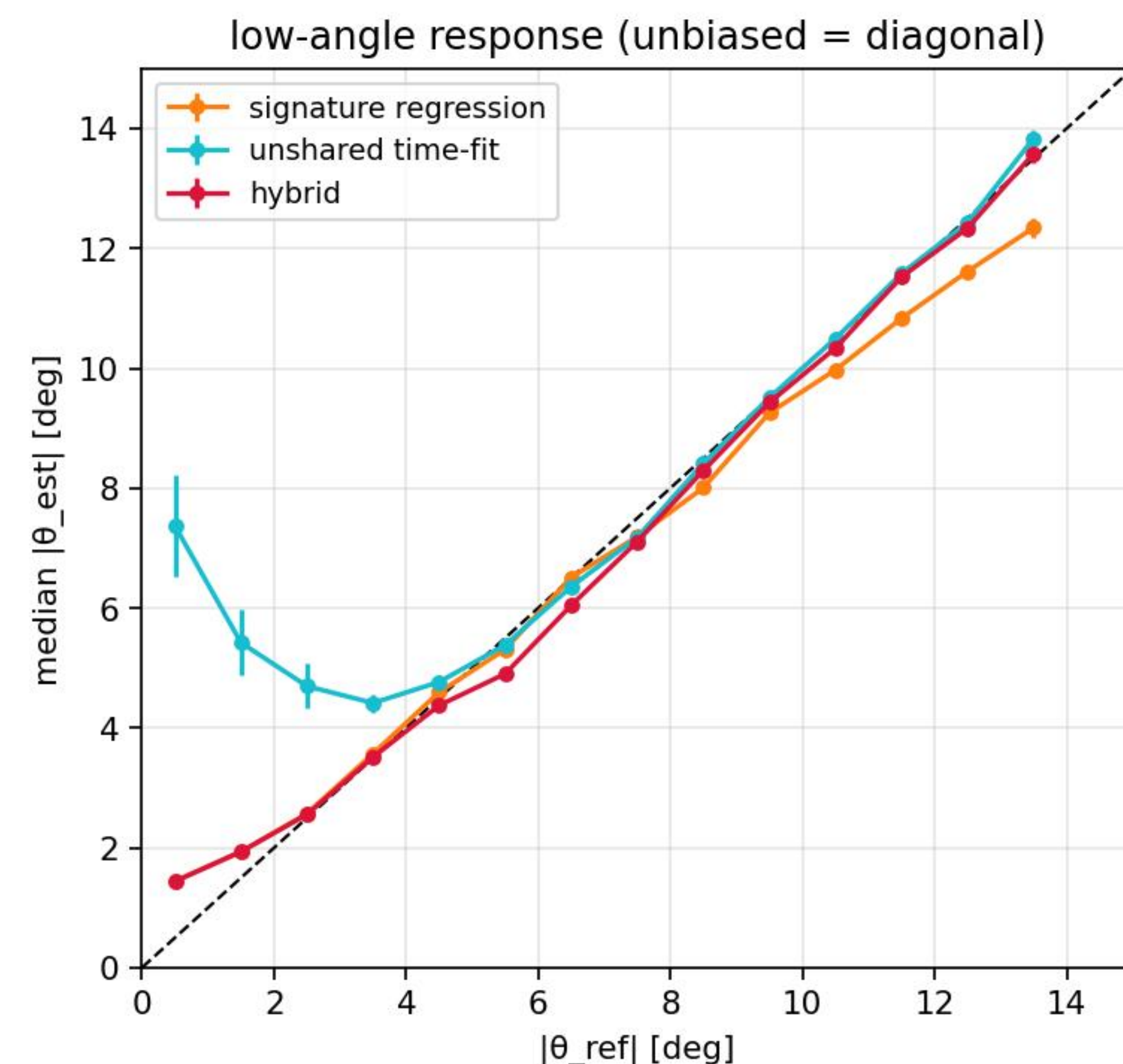
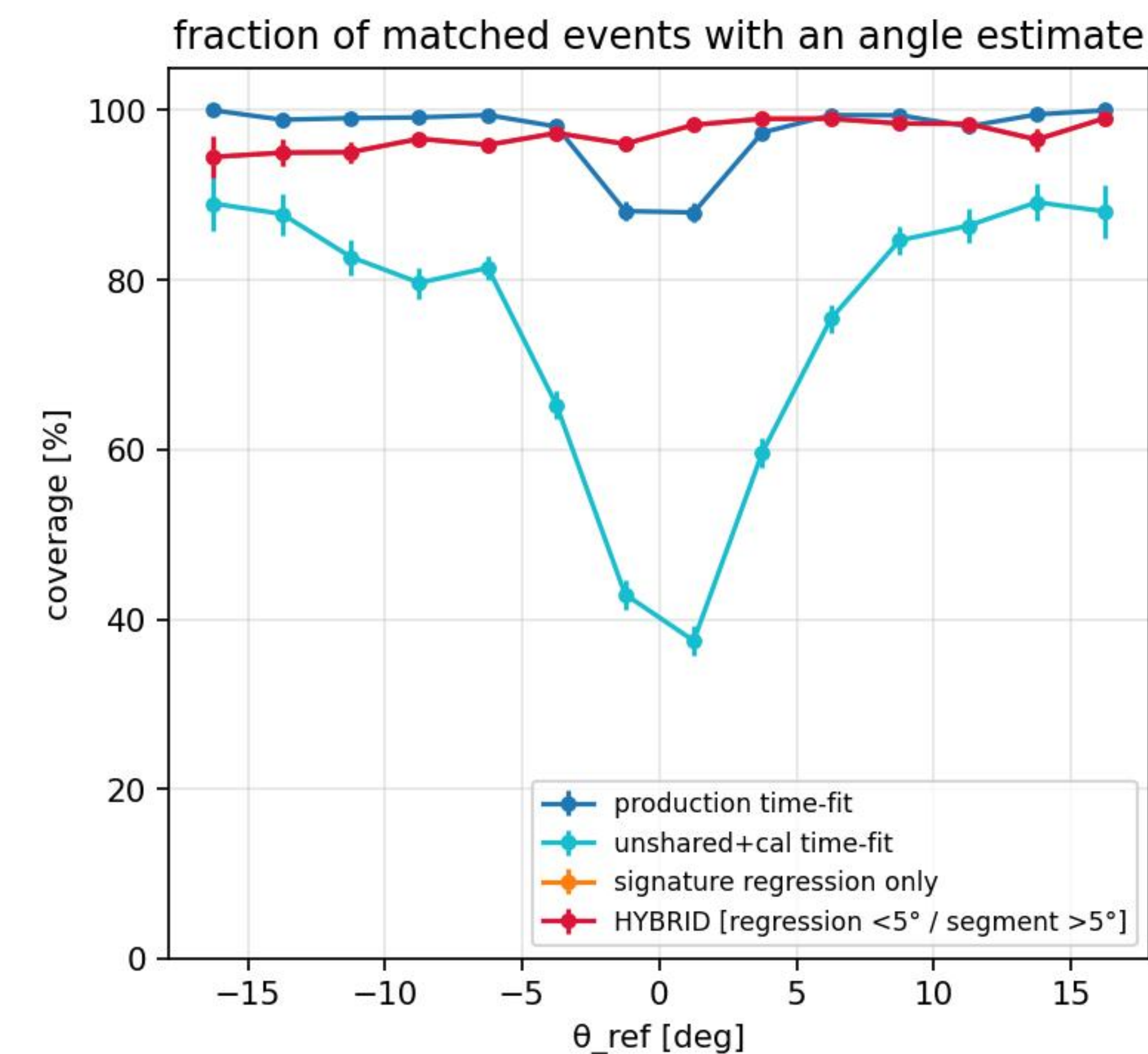
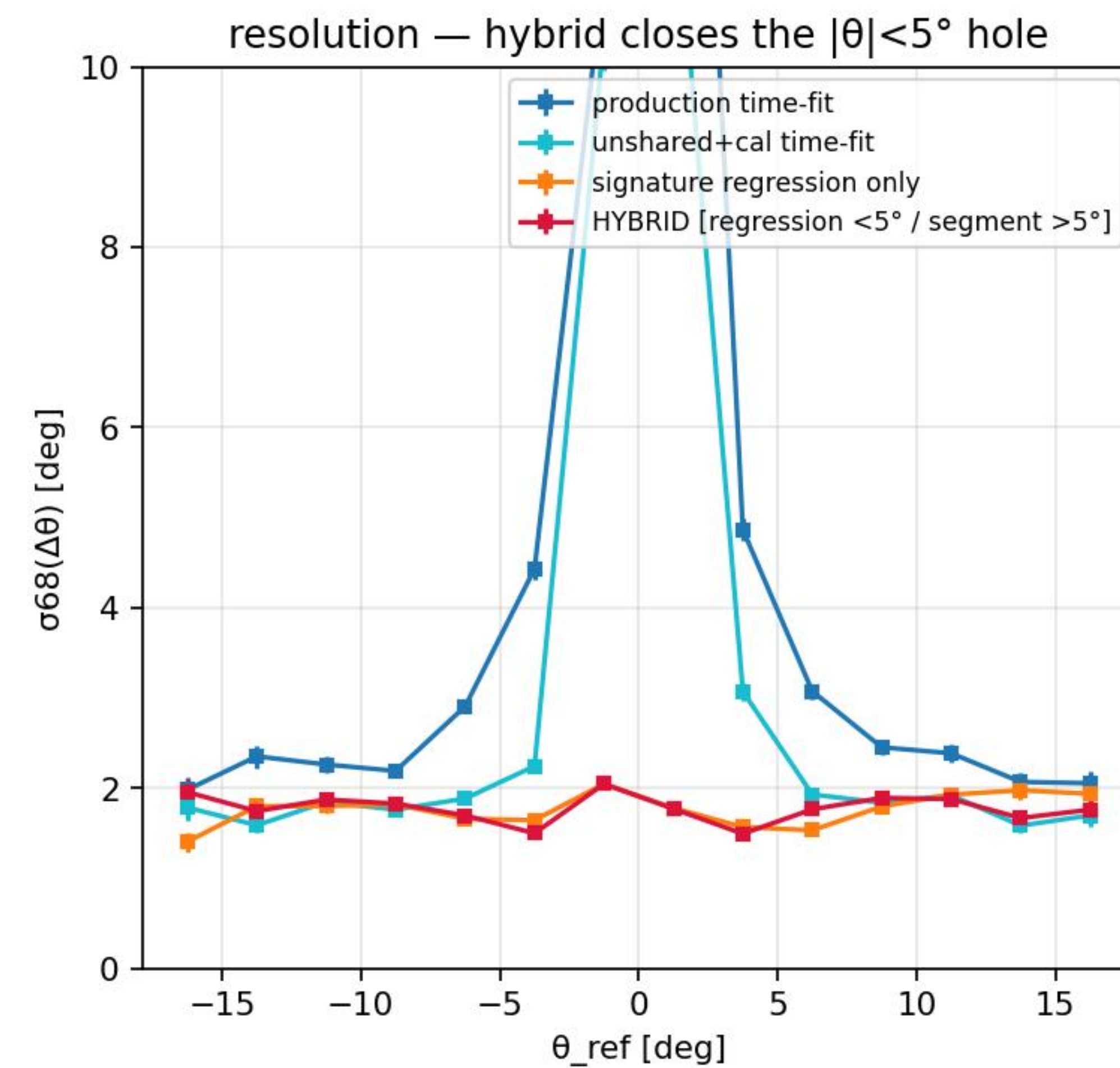
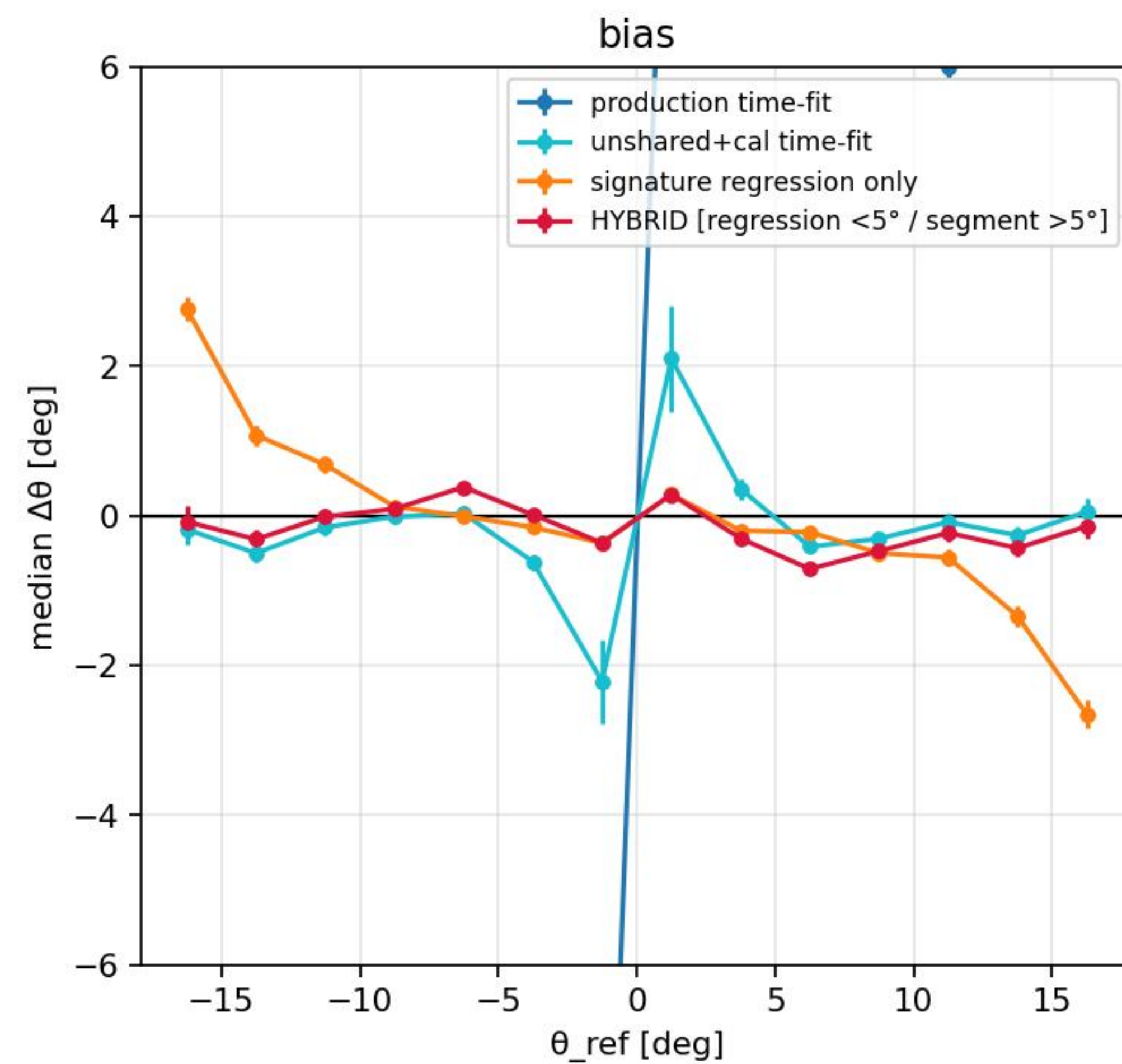
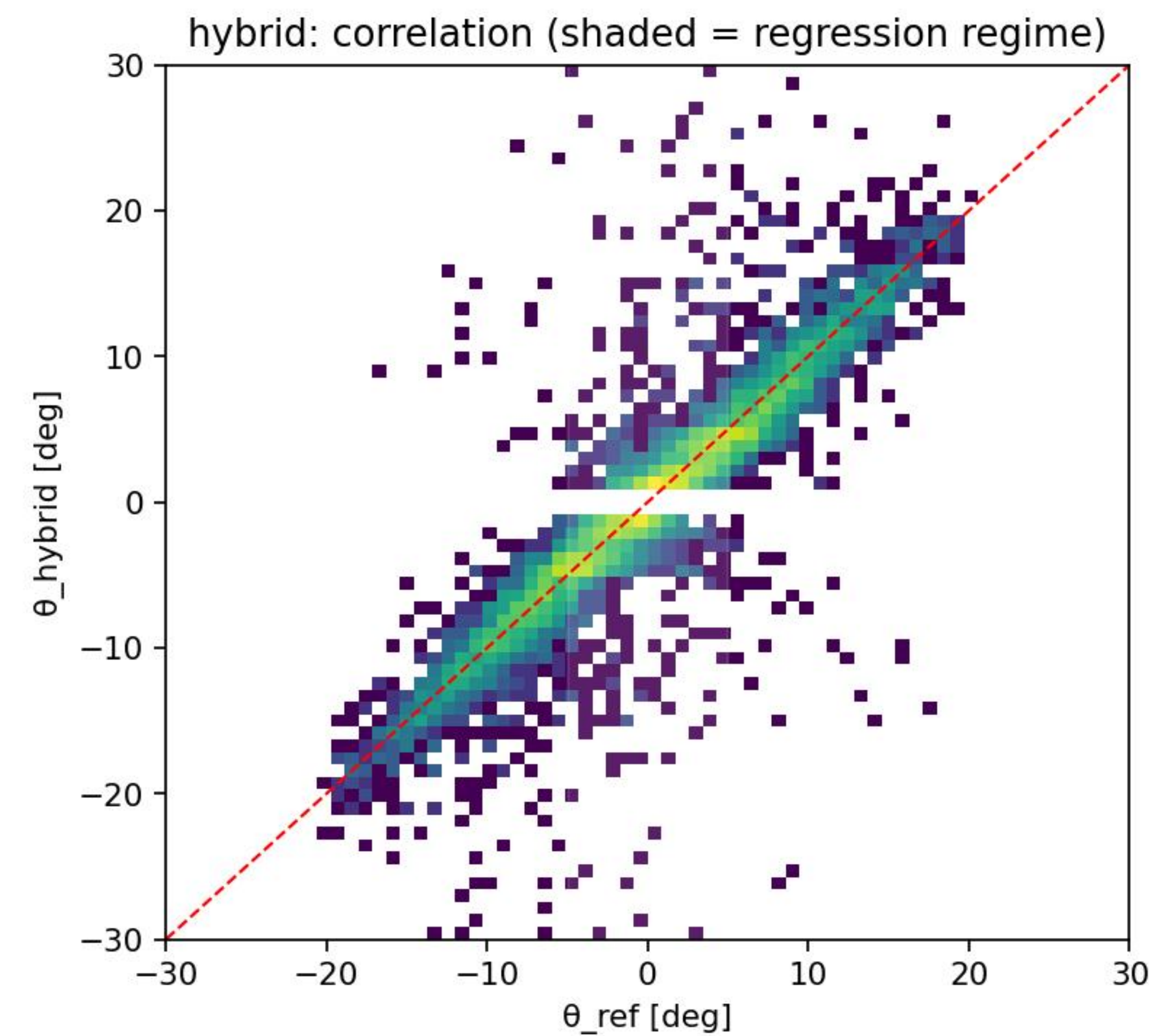




$|\theta_{\text{ref}}| < 5^\circ$ BAND (odd-eid holdout)

estimator	cov	bias	σ_{68}
production time-fit	93%	-3.2°	15.6°
unshared+cal time-fit	51%	-0.2°	5.1°
signature regression only	98%	-0.1°	1.7°
HYBRID [regression $<5^\circ$ / segment $>5^\circ$]	98%	-0.1°	1.7°

sign model (feature-based, x plane):

accuracy 1° :73% 3° :92% 5° :95% 7° :97% 9° :97% 11° :94% 13° :94% 15° :94% 17° :89%

switch: $|\tan_{\text{seg}}| > 0.09$ ($\approx 5^\circ$)

regression features:

tot_lead , q_{frac} , n_u , n_{raw} , a_{asym} , a_{lead} , t_{delay}

Hybrid gives an angle for EVERY event with a lead strip ($\sim 100\%$ coverage) and a finite, calibrated response at $\theta \rightarrow 0$, at the cost of $|\theta|$ -folding noise from the sign model below $\sim 3^\circ$.